

VINTON VETERANS MEMORIAL, IA, USA

WMO: 720326

Lat: 42.219N

Lon: 92.026W

Elev: 845

StdP: 14.25

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04981

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.6	(c) -2.0	(d) -14.3	(e) 2.6	(f) -6.7	(g) -9.3	(h) 3.4	(i) -1.3	(j) 27.3	(k) 23.7	(l) 24.7	(m) 21.0	(n) 6.1	(o) 300	(p) 0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	91.5	74.9	90.4	74.2	87.1	72.4	78.3	88.8	76.4	85.8	74.5	82.4	8.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	137.0	83.5	73.2	127.3	80.9	72.1	122.7	79.9	42.3	87.9	40.3	86.1	38.5	82.3	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.1	(b) 20.4	(c) 18.4	(e) -15.9	(f) 96.6	(g) 5.6	(h) 2.1	(i) -19.9	(j) 98.1	(k) -23.2	(l) 99.4	(m) -26.4	(n) 100.6	(o) -30.5	(p) 102.1		
(4) 24.1	(5) 20.4	(6) 18.4	(7) -16.4	(8) 81.8	(9) 5.2	(10) 1.9	(11) -20.2	(12) 83.1	(13) -23.3	(14) 84.2	(15) -26.2	(16) 85.3	(17) -30.0	(18) 86.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	21.2	24.4	37.4	49.5	61.8	71.3	74.2	71.5	65.1	51.9	38.4	26.6
	DBStd	21.06	12.44	12.64	12.06	9.84	9.07	6.00	5.99	5.11	8.53	9.56	10.36	11.51
	HDD50	3360	892	720	421	126	11	0	0	0	3	91	368	728
	HDD65	6529	1357	1138	857	471	173	13	4	6	99	421	798	1192
	CDD50	3203	0	2	30	112	377	639	749	666	458	149	20	1
	CDD65	896	0	0	2	7	73	202	288	207	103	14	0	0
	CDH74	8582	0	0	16	107	748	1920	2876	1775	998	140	2	0
	CDH80	2860	0	0	1	15	229	658	1094	504	334	25	0	0
	WSAvg	8.1	9.1	9.1	9.7	10.6	8.9	7.2	5.6	5.1	6.3	7.9	9.1	8.6
	PrecAvg	36.9	1.2	1.1	2.2	3.5	4.8	5.7	4.8	4.4	3.2	2.8	2.4	1.1
(15) Precipitation	PrecMax	57.7	2.5	2.6	5.0	7.6	9.3	10.6	12.8	11.5	5.2	6.6	6.2	2.2
	PrecMin	30.8	0.4	0.2	0.2	1.3	0.8	1.5	0.7	1.0	1.0	0.5	0.3	0.2
	PrecStd	6.6	0.5	0.6	1.5	1.6	1.9	2.4	3.0	2.5	1.1	1.8	1.7	0.6
	PrecStd	6.6	0.5	0.6	1.5	1.6	1.9	2.4	3.0	2.5	1.1	1.8	1.7	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	51.2	61.5	76.9	81.8	91.4	95.6	96.1	91.3	93.9	82.5	71.4	57.7	57.7
	0.4% MCWB	43.0	50.0	61.4	60.5	71.4	75.9	78.0	75.7	74.2	65.8	57.7	54.5	54.5
	2% DB	42.9	51.5	68.2	77.2	85.8	90.9	92.6	90.0	90.0	78.0	63.3	50.7	50.7
	2% MCWB	39.0	44.7	56.1	59.3	67.2	73.1	77.0	75.6	72.7	62.3	53.5	46.5	46.5
	5% DB	39.3	44.6	61.0	72.1	81.3	88.3	90.4	85.4	83.4	73.0	58.6	45.3	45.3
	5% MCWB	35.8	39.4	52.9	56.1	64.1	71.7	75.6	73.0	69.0	61.3	50.6	41.8	41.8
	10% DB	36.3	40.6	54.6	67.5	77.6	82.5	86.3	81.8	79.3	69.0	54.3	40.5	40.5
	10% MCWB	34.3	36.4	47.4	54.3	63.5	68.6	73.0	70.5	66.4	59.7	47.6	37.5	37.5
	0.4% WB	43.8	53.0	62.8	64.7	73.2	79.7	81.7	80.4	77.0	71.4	59.3	53.3	53.3
	0.4% MCDB	48.5	62.1	74.3	76.3	88.4	91.0	92.5	88.4	89.2	77.0	66.4	54.9	54.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	39.8	45.4	58.5	62.0	70.6	76.2	79.1	77.4	74.5	67.9	55.8	47.0	47.0
	2% MCDB	43.7	51.2	65.4	73.6	82.2	86.4	90.0	85.7	82.4	74.5	62.9	49.5	49.5
	5% WB	36.5	40.0	53.2	59.0	68.3	73.9	77.0	74.9	72.4	63.7	51.5	41.3	41.3
	5% MCDB	38.5	44.3	58.8	69.3	77.4	82.9	87.6	81.8	80.0	70.1	57.9	45.0	45.0
	10% WB	34.5	36.2	47.9	55.4	66.2	71.8	75.0	73.1	70.1	59.4	47.8	37.2	37.2
	10% MCDB	36.3	39.2	54.3	65.2	74.4	80.4	83.1	79.4	77.0	67.4	53.4	39.4	39.4
(35) Mean Daily Temperature Range	MDBR	15.4	16.4	17.3	20.7	21.4	20.4	20.6	20.4	22.5	20.3	17.5	14.5	14.5
	5% DB	18.1	21.8	25.7	29.3	25.4	25.4	23.3	23.1	25.3	25.7	24.5	19.0	19.0
	5% WB	14.6	15.8	15.8	16.0	11.3	11.5	10.6	11.6	11.1	13.8	16.4	15.2	15.2
	MCDBR	15.9	21.3	22.8	24.2	21.8	21.0	21.4	19.4	20.4	20.4	23.5	18.0	18.0
	MCWBR	13.3	15.8	15.0	15.3	11.6	11.4	10.9	11.2	10.6	14.4	17.5	15.0	15.0
	MCWBR	13.3	15.8	15.0	15.3	11.6	11.4	10.9	11.2	10.6	14.4	17.5	15.0	15.0
(40) Clear-Sky Solar Irradiance	taub	0.284	0.308	0.331	0.377	0.409	0.422	0.435	0.415	0.385	0.338	0.299	0.282	0.282
	taud	2.387	2.329	2.382	2.278	2.257	2.267	2.214	2.279	2.324	2.455	2.488	2.423	2.423
	Ebn at Noon	274	284	290	282	275	271	266	267	268	270	267	265	265
	Edh at Noon	25	31	34	41	43	42	44	40	36	28	23	22	22
(44) All-Sky Solar Radiation	RadAvg	584	908	1211	1485	1761	1944	1964	1714	1428	935	630	467	467
	RadStd	59	78	121	124	123	174	126	109	107	115	51	41	41

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page